FINAL YEAR PROJECT PROPOSAL

Halophyte Salt Concentration Predictor

Project Title	Halophyte Salt Concentration Predictor
Group Members	Ammad Khan - EB-23210006013 Ibadullah Khan - EB-23210006034 Muhammad Masood Sheikh - EB-23210006079 Rohaan Ahmed Khan - EB-23210006112
Supervisor(s)	Dr. Nadeem Mehmood Mukesh Kumar Rathi
Domain Expert	Dr. Mohammad Zaheer Ahmed
Department/University	Department of Computer Science, Umaer Basha Institute of Technology
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1. Abstract

Halophytic grasses can survive in saline environments, but their sodium (Na^+) and potassium (K^+) accumulation varies according to species, salt-tolerance mechanism, tissue type, biomass, growth stage, and environmental conditions. Traditional laboratory measurement of ion concentration is useful and accurate, but it is time-consuming, equipment-dependent, and often limited to individual species or treatments. Published values are also scattered across different research papers, making comparison and estimation difficult. This project proposes a research-paper-based machine learning regression system named Halophyte Salt Concentration Predictor. The system will use extracted values such as grass/species type, salt-secreting or non-salt-secreting behavior, salinity level, tissue type, plant length, water content, chlorophyll content, growth duration, fresh biomass, and dry biomass to predict Na^+ concentration, K^+ concentration, and the Na^+/K^+ ratio. The dataset will be prepared from selected academic research papers, cleaned, standardized, encoded, and used to train regression models such as multiple linear regression and random forest regression. Model performance will be evaluated using R^2, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). The expected outcome is a structured dataset, trained predictive models, a simple prototype notebook, and documentation explaining the important factors affecting ion accumulation in halophytic grasses.

2. Introduction & Background

Halophytes are plants that can grow in saline environments where many normal plants cannot survive. Halophytic grasses and related salt-tolerant plants are important for studying salt tolerance because they manage sodium and potassium balance through mechanisms such as salt storage, exclusion, dilution, and secretion.

Sodium (Na^+) accumulation can cause stress and toxicity in plants, while potassium (K^+) supports plant growth, metabolism, enzyme activity, and water balance. The Na^+/K^+ ratio is commonly used as an indicator of salt-stress tolerance because plants that maintain a healthier potassium balance under salinity usually perform better than plants that accumulate excessive sodium.

The specific problem addressed in this project is the difficulty of estimating Na^+ and K^+ concentration from multiple plant, biomass, physiological, and environmental factors without performing new laboratory tests. Existing studies often report useful values, but those values are usually distributed across separate papers, species, treatments, and measurement conditions.

This project will therefore use machine learning regression to predict Na^+ concentration, K^+ concentration, and Na^+/K^+ ratio from research-paper-extracted values. The project does not aim to replace laboratory testing. Instead, it aims to provide an estimated prediction and comparative analytical tool based on available literature data.

3. Literature Review / Related Work

Khan studied the germination response of Suaeda fruticosa from Pakistan under different salinity and temperature conditions [1]. The study is relevant because it shows how salt-tolerant plants respond differently under changing environmental conditions. However, the focus of the work was mainly on germination behavior rather than predicting Na^+ or K^+ concentration using combined plant parameters.

Gulzar and Khan investigated diurnal water relations of inland and coastal halophytic populations from Pakistan [2]. Their work provides useful biological context about how halophytes manage water relations in saline environments. However, it does not combine physiological, biomass, and ion-related variables into a machine learning prediction system.

Khan and Ungar analyzed seed germination and dormancy of Polygonum aviculare L. as influenced by salinity, temperature, and gibberellic acid [3]. This work supports the importance of salinity and environmental conditions in plant response studies. However, the study is experimental and species-specific, and it does not provide a generalized computational model for ion concentration estimation.

Gul and Khan reported population characteristics of the coastal halophyte Arthrocnemum macrostachyum [4]. The study is useful for understanding species-specific characteristics of coastal halophytes, but it does not provide a predictive model that can estimate Na+ concentration, K+ concentration, or Na+/K+ ratio from extracted features.

Overall, existing research provides valuable species-level and treatment-level data, but most studies report results separately for specific halophytes, locations, salinity levels, or growth conditions. The identified research gap is the lack of a structured machine learning workflow that combines research-paper-extracted halophyte data into a regression-based prediction system for Na+ concentration, K+ concentration, and Na+/K+ ratio. This project fills that gap by preparing a structured dataset and comparing regression models for literature-based ion concentration prediction.

4. Aims and Objectives

General Objective

To develop a machine learning regression system that predicts Na+ concentration, K+ concentration, and Na+/K+ ratio in halophytic grasses using research-paper-extracted biomass, physiological, and environmental parameters.

Specific Objectives

- To review at least 3-5 relevant academic research papers related to halophytes, salinity response, biomass, and ion accumulation.
- To extract and organize available values such as species type, salinity level, tissue type, growth duration, chlorophyll content, water content, fresh biomass, dry biomass, Na+ concentration, and K+ concentration into a structured dataset.
- To clean the dataset by handling missing values, standardizing measurement units, and encoding categorical variables for machine learning use.
- To train and compare at least two regression models, including multiple linear regression and random forest regression, for predicting Na+ and K+ concentration.
- To evaluate model performance using R^2, MAE, and RMSE, and develop a simple notebook-based prototype for entering parameters and generating predicted outputs.

5. Project Scope

What the Project Will Do

- Use data extracted from published academic research papers only.
- Prepare a structured dataset for halophyte-related biomass, physiological, environmental, and ion concentration values.
- Predict Na+ concentration, K+ concentration, and Na+/K+ ratio as numeric outputs.

- Use input features such as grass/species type, salinity level, tissue type, plant length, water content, chlorophyll content, growth duration, fresh biomass, and dry biomass where available.
- Compare at least two regression models, such as multiple linear regression and random forest regression.
- Produce a trained model, dataset, evaluation results, prototype notebook, final documentation, and presentation.

What the Project Will Not Do

- It will not conduct new laboratory experiments during the current FYP phase.
- It will not use direct laboratory measurement equipment because values will be extracted from published research papers.
- It will not claim to replace laboratory testing; it will provide estimated predictions based on available literature data.
- It will not guarantee high accuracy if the extracted dataset is too small, inconsistent, or missing important variables.
- Final grass/species selection will depend on literature review results and dataset availability.

6. Proposed Methodology & Design

System Architecture

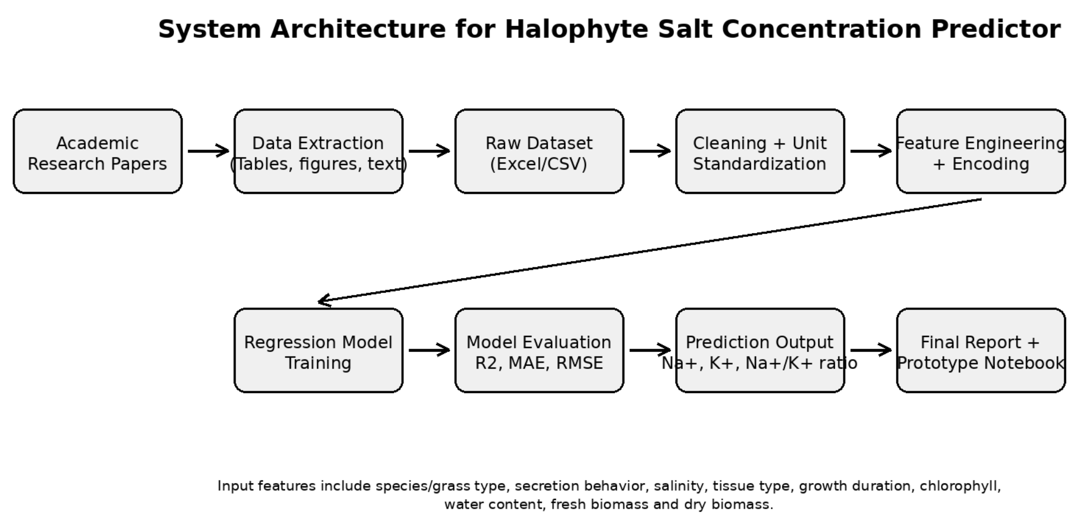

Figure 1. Proposed system architecture for the Halophyte Salt Concentration Predictor.

Methodology

Paper Selection: Relevant academic papers on halophytes, salinity response, biomass, water content, chlorophyll, and ion concentration will be selected.

Data Extraction: Numerical and categorical values will be extracted from tables, figures, and text where available. Extracted values will be entered into Excel/CSV format.

Dataset Preparation: The dataset will be organized with clear feature columns, target columns, units, source-paper references, and notes for missing or estimated values.

Data Cleaning and Standardization: Missing values will be handled carefully, units will be standardized, duplicate records will be removed, and categorical variables will be encoded.

Model Training: Multiple linear regression will be used as a baseline model, and random forest regression will be used to capture possible non-linear relationships between plant parameters and ion concentration [5].

Model Evaluation: The models will be evaluated using R^2, MAE, and RMSE. If the dataset size allows, train-test splitting or cross-validation will be used.

Prediction Prototype: A simple Jupyter Notebook or Google Colab prototype will allow users to enter available parameters and obtain predicted Na^+ concentration, K^+ concentration, and Na^+/K^+ ratio.

Main Input Parameters

Parameter	Use in Model
Grass/species type	Categorical input representing the halophyte species or grass type.
Salt-secreting or non-salt-secreting type	Categorical input describing the salt-tolerance mechanism where available.
Salinity level / environmental condition	Numeric or categorical input based on the treatment condition reported in the paper.
Plant/tissue type	Root, shoot, leaf, or whole plant depending on available data.
Plant length	Numeric input if reported in the selected paper.
Water content	Percentage or extracted value where available.
Chlorophyll content	SPAD, lab-measured, or extracted value where available.
Growth duration	Days, weeks, or growth stage as reported in the paper.
Fresh biomass and dry biomass	Central numeric input features for predicting ion accumulation.

Outputs

- Predicted Na^+ concentration
- Predicted K^+ concentration
- Calculated or predicted Na^+/K^+ ratio
- Model performance summary using R^2, MAE, and RMSE

Tools & Technologies

- Python for implementation
- Pandas and NumPy for data handling and preprocessing
- Scikit-learn for machine learning model training and evaluation [6]
- Matplotlib for visualizing results where required
- Excel/CSV files for dataset preparation
- Jupyter Notebook or Google Colab for experimentation and prototype development

7. Expected Outcomes & Deliverables

- A structured dataset prepared from research-paper-extracted halophyte values.
- A cleaned and standardized dataset ready for machine learning use.
- Trained regression models for Na+ and K+ concentration prediction.
- Na+/K+ ratio calculation and analysis.
- Model evaluation results using R^2, MAE, and RMSE.
- A simple notebook/prototype predictor for entering parameters and generating predicted values.
- Final FYP report, presentation slides, and supporting documentation.

8. Timeline / Gantt Chart

Project duration: 2 months

Task	W1	W2	W3	W4	W5	W6	W7	W8
Finalize topic & papers	X	X						
Literature review & extraction		X	X					
Clean & standardize data			X	X				
Feature engineering				X	X			
Model training				X	X			
Model testing & evaluation					X	X		
Prototype notebook						X	X	
Report & presentation							X	X

9. References

[1] M. A. Khan, "Germination of the salt tolerant shrub Suaeda fruticosa from Pakistan: salinity and temperature responses," Seed Science and Technology, vol. 26, no. 3, pp. 657-667, 1998.

[2] S. Gulzar and M. A. Khan, "Diurnal water relations of inland and coastal halophytic populations from Pakistan," Journal of Arid Environments, vol. 40, no. 3, pp. 295-305, 1998.

[3] M. A. Khan and I. A. Ungar, "Seed germination and dormancy of Polygonum aviculare L. as influenced by salinity, temperature, and gibberellic acid," Seed Science and Technology, vol. 26, pp. 107-117, 1998.

[4] B. Gul and M. A. Khan, "Population characteristics of the coastal halophyte Arthrocnemum macrostachyum," Pakistan Journal of Botany, vol. 30, no. 2, pp. 189-197, 1998.

[5] L. Breiman, "Random forests," Machine Learning, vol. 45, pp. 5-32, 2001, doi: 10.1023/A:1010933404324.

[6] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.

10. Signatures

Name	Role	Signature
Ammad Khan	Student	
Ibadullah Khan	Student	
Muhammad Masood Sheikh	Student	
Rohaan Ahmed Khan	Student	
Supervisor	Approval	